

Arghya Datta, PhD

Applied Scientist
Worldwide Amazon Stores (WWAS)
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Educational Qualification

PhD in Computer Science, [Washington University in St. Louis](#), Department of Computer Science & Engineering, St. Louis, USA

MS in Computer Science, [Washington University in St. Louis](#), Department of Computer Science & Engineering, St. Louis, USA

Bachelor of Technology, [West Bengal University of Technology](#), Department of Electronics & Communication Engineering, Kolkata, India

Work Experience

Company	Amazon.com Inc
Tenure	Sept 2021- Present
Role	Applied Scientist, Applied AI (eCF)., WorldWide Retail Stores
Work Summary	Lead scientist for building AgentZ, an agentic LLM framework for automating knowledge operations work across worldwide retail stores. - Led conceptualization and development of SOP-Bench, a SOTA LLM benchmark generation framework to text LLM's reasoning, planning and tool calling capabilities for operationalizing standard operating procedures (SOPs). - Built a LLM based text-to-sql engine to power Amazon's business intelligence use cases that supported customers querying an LLM to extract information from relational databases. The system was built on RAGs, Chain-of-Thought (CoT), self-correction, self-consistency techniques and achieved an execution accuracy of 97% for writing simple SQL queries. The system achieved a rank of 10 on the SPIDER benchmark Leaderboard. - Built a LLM based annotation agentic workflow that generated annotated training data using a HITL approach for the Prime Air's use cases pertaining to training large scale vision algorithms for Mid-Air Collision Detection Systems. The framework resulted in an 85% efficiency (cost reduction) for generating training data for ground object detection resulting in savings greater than \$1.7MM.
Role	Applied Scientist, Amazon Q Developer (NGDE), Amazon Web Services
Work Summary	Leading science efforts for the development of Amazon Q Code Developer Agent, Amazon's flagship GenAI product for helping developer generate and transform code using Large Language Models.Company Announcement - Led development of code transformation platform that migrated JDK packages saving 4500 developer years - Led development of code transformations for packages from AwsSdk v1 to v2, AWS's most used Java SDK platform
Role	Applied Scientist, Artificial General Intelligence Org.
Work Summary	Lead Scientist working on fine-tuning, model performance enrichment & model robustness - Designed a self learning approach that iteratively identifies gaps in model's capabilities and identifies high quality training sets for fine-tuning. - Invented and deployed a neural network-based model to reduce instability of large language models powering Amazon Alexa by 35%. Work published in Association for Computational Linguistic Conference 2023 - Built Amazon's core capabilities for Large Language Model's data quality framework for training large language models (Amazon Titan Model family) as well as selection of datasets to bridge model gaps.

Role	Applied Scientist, Customer Targeting, WorldWide Retail Stores
Work Summary	Conceptualized and deployed Amazon's customer action ranking models as well as MAB-based autonomous targeting systems - Developed a novel customer ranking algorithm based on neural networks that is currently deployed in all worldwide markets by Amazon's Customer Targeting Team to target retail customers for shopping on Amazon. Work published in Amazon Machine Learning Conference; A/B testing showed downstream profit of \$250,000 - Developed multi-arm bandit based Objective Based Targeting System that is used by Amazon worldwide to automate marketing campaigns to drive targeting to satisfy a business objective Deployment showed an increment of 8% across campaigns on electronics, beauty, toys - Developed Amazon's Customer Segmentation Insights tool that leverage statistical machine learning to identify key characteristics of customer shopping trends that affect contributing profit.

Previous Affiliations

Jun 2019- Sep 2019	Data Scientist Intern, Amazon.com Inc , World-Wide Amazon Stores Organization, Seattle, USA
August 2016- August 2021	Doctoral Researcher, Washington University in St. Louis , Department of Computer Science & Engineering, St. Louis, USA
Aug 2015- May 2016	Software Engineer, Infosys Limited , Pune, India

Publications

Google Scholar [Arghya Datta](#)

Preprints

The following articles are under review and published on *Arxiv*

- Subhrangshu Nandi†, **Arghya Datta†**, Nikhil Vichare†, Indranil Bhattacharya†, Huzefa Raja, Jing Xu, Shayan Ray, Giuseppe Carenini, Abhi Srivastava, Aaron Chan, Man Ho Woo, Amar Kandola, Brandon Theresa, Francesco Carbone *SOP-Bench: Complex Industrial SOPs for Evaluating LLM Agents*, Arxiv, 2023 († denotes equal contribution, co-first authors)

Peer-Reviewed Articles

The following articles are published in peer-reviewed premier journal & conferences.

- Arghya Datta**, Subhrangshu Nandi, Jingcheng Xu, Greg Ver Steeg, He Xie, Anoop Kumar, Aram Galstyan, *Measuring and Mitigating Local Instability in Deep Neural Networks*, Findings for Association of Computational Linguistics (ACL), 2023
- Arghya Datta**, SJ Swamidass. *Fair-Net: A Network Architecture for Reducing Performance Disparity between Identifiable Sub-populations*, In Proceedings of the 14th International Conference on Agents and Artificial Intelligence (ICAART), 2022
- Arghya Datta**, Noah R Flynn, SJ Swamidass. *Cal-net: Jointly learning classification and calibration on imbalanced binary classification tasks*, International Joint Conference on Neural Networks (IJCNN), 2021
- Arghya Datta**, Noah R Flynn, Dustyn A Barnette, Keith F Woeltje, Grover P Miller, S Joshua Swamidass. *Machine learning liver-injuring drug interactions with non-steroidal anti-inflammatory drugs (NSAIDs) from a retrospective electronic health record (EHR) cohort*, PLOS Computational Biology 2021
- Dustyn Barnette, Mary Schleiff, , **Arghya Datta**, Noah Flynn, S Joshua Swamidass, Grover P. Miller. *Meloxicam methyl group determines enzyme specificity for thiazole bioactivation compared to sudoxicam*, Toxicology Letters 2021
- Arghya Datta**, Matthew K Matlock, Na Le Dang, Thiago Moulin, Keith F. Woeltje, Elizabeth L. Yanik, Sanjay Joshua Swamidass. *Black Box to Conversational Machine Learning: Ondansetron Reduces Risk of Hospital-Acquired Venous Thromboembolism*, IEEE journal of Biomedical and Health Informatics 2020

- Matthew K Matlock, **Arghya Datta**, Na Le Dang, Kevin Jiang, S Joshua Swamidass. *Deep learning long-range information in undirected graphs with wave networks*, International Joint Conference on Neural Networks (IJCNN), 2019

**Amazon
Machine
Learning
Conference
[Internal]**

The following articles are published in peer-reviewed internal Amazon Machine Learning Conference (acceptance rate < 15%), which is attended by top academics from all over the world

- **Arghya Datta**, Pragyana Mishra, *Ranking Customer Actions using Propensity Calibration in Neural Networks*, Amazon Machine Learning Conference (AMLC), 2022
- **Arghya Datta**, Subhrangshu Nandi, Anoop Kumar, Aram Galstyan, Greg Ver Steeg, Abhishek Kumar, Purva Kulkarni. *Measuring and Mitigating Local Instability in Large Language Models*, Amazon Machine Learning Conference (AMLC), 2023

Academic, industry & volunteer service

Peer reviewer Served as a reviewer for more than **140+** reviews for renowned international Journals and Conferences.

Journals Served as the key peer-reviewer for the following journals

- Journal of Biomedical and Health Informatics Journal (IEEE)
- Transactions on Computational Social Systems Journal (IEEE)
- Signal Processing Letters (IEEE)
- Applied AI Letters (Wiley)
- Frontiers in Neurorobotics
- Frontiers Energy Research
- Frontiers Ecology and Evolution
- Frontiers Public Health
- Journal of Cellular and Molecular Medicine (Wiley)
- IET Computer Vision (Wiley)
- Cancer Reports (Wiley)
- IET Image Processing (Wiley)
- Journal of Mathematics (MDPI)
- Journal of Healthcare (MDPI)
- Journal of Electronics (MDPI)
- Journal of Applied Sciences (MDPI)
- International Journal of Environmental research and public health (MDPI)
- Journal of BioMedInformatics (MDPI)

Conferences Served as the key peer-reviewer for the following conferences

- International Conference on Artificial Intelligence and Statistics (AISTATS) conference
- Amazon Machine Learning Conference (AMLC)

Session Chair Served as the session chair for key research topics in deep learning

- International conference on Agents and Artificial Intelligence (ICAART), 2022 for topic on **Deep Learning**.
- International conference on Agents and Artificial Intelligence ICAART), 2022 for topic on **Natural Language Processing**.
- International conference on Agents and Artificial Intelligence ICAART), 2022 for topic on **AI and creativity**.
- Amazon Machine Learning Conference (AMLC) 2022 for Workshop on **Robust Machine Learning**.

Invited Talks

Oral Presentations

Was invited to present research at the following venues and topics

- International Joint Conference on Neural Networks (IJCNN) at Budapest, Hungary in 2019 titled *Deep learning long-range information in undirected graphs with wave networks*.
- International Joint Conference on Neural Networks (IJCNN) in 2021 titled *Classification and Calibration in data scarce scenarios for high-risk applications*.
- International Conference on Agents and Artificial Intelligence (ICAART) 2022 titled *Fairness and calibration in class imbalanced scenarios*.
- Amazon Machine Learning Conference (AMLC) 2022 titled *Ranking Customer Actions for efficient targeting*.
- Amazon Machine Learning Conference (AMLC) 2023 titled *On Stability of Large Language Models*.

Research Interest

- Deep learning
- Neural Networks
- Knowledge discovery using Machine Learning
- Graph Neural Networks
- Large Language Models (LLMs)
- Natural Language Understanding
- Time Series Forecasting
- Causal Modelling
- Object detection and Segmentation
- Personalization, Recommendation Systems and Targeting
- Advanced Statistics, A/B testing

Technical Skills

Coding Languages	C, C++, Java, Python, JavaScript, Matlab, UNIX shell scripting, AWK programming
RDBMS	Expertise in handling databases implemented using Oracle/MS SQL via advanced Structured Query Language(SQL)
Deep Learning	Pytorch, Tensorflow and Keras
NLP/Image Processing	Expertise in optimizing NLP algorithms and using popular python libraries like OpenCV(Used for numerous side projects)
Complex Data Handling	Expertise in Hadoop, Spark, Hive, Dask, Pandas, Parquet file system, Tempfiles as well as HDF5 including major python libraries

Teaching & Mentoring Experience

Coursework	Taught a course titled Intro to Machine Learning with the largest enrollment of 170 students at WashU and conducted labs, evaluated homework and exam papers
Research Associate	Mentored students at Swamidass Lab: Graph deep learning paper, a novel research work, was published at IJCNN 2019.
REU	Mentored WashU undergrads in Swamidass Lab in Machine Learning Algorithms and writing libraries for chemical data processing